

JAZZ MEDIA CORES

MOBILE MEDIA PROCESSORS



Mobile Phones



Portable Music Players



Portable Media Players



Web Cameras



Mobile IP TV

Low power, high performance cores that enable rapid support of robust media processing.

The **Jazz Mobile Media Processor (MMP) cores** are highly optimized solutions specifically tailored to the demanding needs of the mobile and portable media market. Whether you are replacing an existing solution or adding media processing to your designs, the Jazz MMP cores can get you to market quickly and efficiently. They support a wide range of video, audio, image and speech standards and are optimized for low power, low cost applications. The MMP cores are part of the broad Jazz Media family from Improv.

New portable media devices require support for a variety of different video, audio, image and speech standards. The Jazz MMP solutions support a complete set of media software modules that can be loaded from external memory; this allows a single core to support a broad range of media standards.

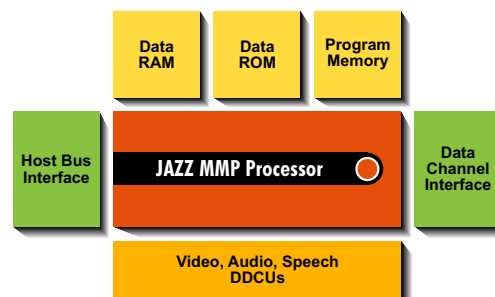
The Jazz MMP cores provide a complete package for supporting a wide variety of broadcast quality media standards in your product. The cores are pre-configured and pre-optimized and come with a complete suite of media software modules. The cores have built-in interfaces for standard on-chip buses and memory management to allow direct drop-in into an existing SoC architecture.

OPTIMIZED CORES

The Jazz MMP cores can be used as a black box implementation of complete elementary stream processing for a mobile/portable media chip design. Each core can operate as a standalone block or as a slave to a host processor.

Each MMP core has a single, application-specific processor that combine a general-purpose instruction set with application-specific instructions that accelerate media processing functions. The processors are very-long instruction word (VLIW) processors built using Improv's Jazz Composer tools.

In addition to the processor, the hardware core contains embedded memory, a host bus interface (HBI) that can connect to most standard on-chip buses including AMBA, OCP and Wishbone, and a media-aware memory interface called the Data Channel Interface (DCI).



KEY FEATURES

- Complete silicon-ready solution
- Optimized for small die size and low power
- Low clock rates for advanced media standards
- Available in audio, video and media configurations
- Reduces CPU load by 75-90%.
- Fully synthesizable and foundry independent
- Supported by a comprehensive software development tool suite
- Optionally available as configurable cores for additional customization

MMP MEDIA SOFTWARE MODULES

The MMP media software modules have been designed to fully exploit the unique advantages of the MMP architecture. The MMP family includes the following modules:

- H.264 AVC Baseline Video Decoder
- MPEG4 Simple Profile Codec
- Window Media 9 Video Decoder
- MPEG L1/L2 Audio Decoder
- MP3 Audio Decoder
- AAC Audio Decoder
- MPEG L1/L2 Audio Encoder
- MP3 Audio Encoder
- BSAC Audio Decoder
- Windows Media 9 Audio Decoder
- JPEG Still Image Codec
- G.711, G.726, G.723.1, G.729a Codecs

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AVAILABLE PRODUCTS

The Jazz MMP family consists of the following specific products:

- **MMP Video Decoder Core** is optimized for video decoding and includes all MMP video/image decoding and image modules.
- **MMP Video Codec Core** is optimized for a full-duplex video codec and includes all MMP video/image decoding and codec modules.
- **MMP Media Decoder Core** is optimized for full media decoding and includes all media software modules.
- **MMP Media Codec Core** is optimized for a full-duplex media codec and includes all media software modules.

IMPLEMENTATION CHARACTERISTICS

	<u>GATE COUNT</u>	<u>CLOCK</u>	<u>POWER</u>
MMP Video Decoder	354K gates	50 MHz	23 mW
MMP Video Codec	367K gates	75 MHz	33 mW
MMP Media Decoder	386K gates	75 MHz	35 mW
MMP Media Codec	399K gates	125MHz	60 mW

These numbers represent the maximum requirements for running the MMP Media Software modules in a TSMC 0.13um process.

Power dissipation is a critical component of a successful chip for these markets. The Jazz MMP cores come with sophisticated power management features. In addition to the internal power management built into the cores, the cores support IDLE and SLEEP modes as well as power savings mode to run at lower frame rates.

- Fully synchronous design
- Silicon-proven in 0.18um and 0.13um technologies
- Advanced power management features
- Hardware Debug Environment through JTAG port
- Support for prioritized interrupts
- Full scan with BIST for test
- Boot-loadable firmware and smart caching (Jazz Jukebox)
- No custom physical components required

DEVELOPMENT TOOLS

A complete software development environment called the Jazz Standard tool suite supports Jazz cores. This tool suite includes compiler, debugger, assembler, cycle-accurate instruction-set simulator and profiler/analyzer. The environment runs on Windows XP Professional, Windows 2000 and Windows NT environments.

The cores also come with a complete implementation environment called the Jazz Generator which creates complete synthesizable RTL, EDA tool scripts, verification environment and FPGA mapping.

As a separately purchasable item, Improv offers a development platform using Improv's FPGA-Based Jazz Rehearsal™ cards with a full verification environment to enable verification and system integration.

DELIVERABLES

- MMP Processor Platform
- MMP Media Software Modules
- Verification Environment
- Jazz Standard Tool Suite
- Jazz PSA Generator
- User's Guide
- Implementation Guide
- Programmer's Guide
- Jazz Composer Tool Suite (for configurable package only)

LICENSING INFORMATION

The Jazz MMP cores are available today and can be licensed from Improv Systems. For additional information regarding the Jazz Media family and other licensable Improv technology, please contact Improv Systems at sales@improvsys.com, or visit our website at <http://www.improvsys.com>. A list of worldwide local representatives can be found on the website.

ABOUT IMPROV SYSTEMS, INC.

Improv Systems, Inc. develops and licenses pre-configured and configurable Jazz DSP, Jazz Media and Jazz Voice cores. Improv's Jazz processor is a configurable DSP that allows designers to create optimized DSP cores targeted to consumer electronics and telecommunications markets. For more information, please visit www.improvsys.com.

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